

① Image-Question Dataset

Candidate Generation

Image-Question



Prompt:
How many families?



Candidate
Responses

Response A

The image shows a union organization display.

Response B

The image shows a Union Organization table setup with 18,000 families.

Response C

The image does not provide any information about families.

Decompose each response into visual claims(B example)

Claim 1: union-related evidence

Union-related signs or logos are visible.

Claim 2: family-related evidence

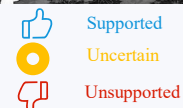
Text related to families is visible.

Claim 3: numerical claim

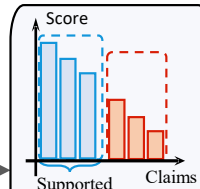
The exact number '18,000 families' is visually supported.

② Visual Evidence Calibration

Evidence checking
against image I



Evidence Score
 $E(y, I)$



Average support of claims in response

③ Evidence-Calibrated Pair Construction



Prompt:
How many families?

Chosen(y_ω)

👍 The image shows a Union Organization table setup with 18,000 families.

Rejected(y_l)

👎 The image does not provide any information about families.

Evidence Gap

$$\Delta E =$$

$$E(y_\omega, I) - E(y_l, I)$$

If > 0 , indicates y_ω is better

Evidence Gap

Chosen(y_ω)

Rejected(y_l)

VEC-DPO

$$\mathcal{L}_{VEC-DPO} = \omega_e \cdot \mathcal{L}_{DPO}$$

$$\omega_e = 1 + \alpha \cdot \Delta E$$

④ VEC-DPO Training

MLLM

Training

**Calibrated
MLLM**

